WAHA (WEBJS Engine) Setup on Windows 11 + Docker

WhatsApp HTTP API for the **EasySchool Management System** | Engine: WEBJS Generated 2026-06-12

> **Maqsad (Purpose):** Yeh guide aapko Windows 11 par Docker ke through WAHA ko **WEBJS engine** ke sath chalana sikhati hai, phir EasySchool API ko us se connect karna. Aapki app already `/api/sendText` aur `/api/sendFile` endpoints, `X-Api-Key` header, session `default` aur country code `92` use karti hai — yeh setup usi ke mutabiq hai.

1. WAHA aur WEBJS engine kya hai?

WAHA (WhatsApp HTTP API) ek self-hosted server hai jo WhatsApp ko ek simple REST API mein convert kar deta hai. Aap HTTP request bhej kar messages/files send kar sakte hain — koi official WhatsApp Business API ki zaroorat nahi.

- **WEBJS** — WhatsApp Web ko emulate karta hai (whatsapp-web.js library). Free, QR-code login. EasySchool ke liye recommended.
- **NOWEB** — halka engine, kam RAM. (Alternative)

2. Prerequisites (Pehle se zaroori cheezein)

Requirement	Detail
Windows 11	Pro / Home (64-bit), virtualization BIOS mein ON
Docker Desktop	Aapne already install kar liya hai ✓ — WSL2 backend enabled hona chahiye
RAM	Minimum 4 GB free (WEBJS Chromium chalata hai)
A WhatsApp number	Jis se messages bhejne hain — uska phone QR scan ke liye paas ho
Internet	Docker image pull karne ke liye

Docker check: PowerShell kholain aur confirm karein Docker chal raha hai:

```
docker --version
docker info
```

Agar error aaye to Docker Desktop app open karke "Engine running" hone ka intezar karein.

3. WAHA Docker image pull karein

Do versions hain:

Image	Note
`devlikeapro/waha`	**Core (free)** — NOWEB + WEBJS engine, text/file bhejne ke liye kaafi. EasySchool ke liye yehi use karein.
`devlikeapro/waha-plus`	Plus (paid) — extra features (groups bulk, media history). License chahiye.

```
# PowerShell mein:
docker pull devlikeapro/waha
```

4. WAHA container chalayein (WEBJS engine ke sath)

Yeh sabse important command hai. `WHATSAPP_DEFAULT_ENGINE=WEBJS` se WEBJS engine force hoti hai, aur `WAHA_API_KEY` aapki app ke `X-Api-Key` se match hona chahiye.

```
docker run -d `
  --name waha `
  --restart always `
  -p 3000:3000 `
  -e WHATSAPP_DEFAULT_ENGINE=WEBJS `
  -e WAHA_API_KEY=4c4662668dff4e48997492b11d4e7754 `
  -e WHATSAPP_HOOK_EVENTS=session.status `
  -e WAHA_PRINT_QR=false `
  -v waha_sessions:/app/.sessions `
  devlikeapro/waha
```

Command samjhein:

- `-d` — background mein chalayega
- `--restart always` — PC restart hone par container khud start ho jayega
- `-p 3000:3000` — WAHA `http://localhost:3000` par milega
- `WHATSAPP_DEFAULT_ENGINE=WEBJS` — WEBJS engine
- `WAHA_API_KEY=...` — yeh wahi key hai jo aapki `appsettings.json` mein hai. Iske bina API protected nahi hogi.
- `-v waha_sessions:/app/.sessions` — session data save rahega taa-ke har restart par dobara QR scan na karna pade

PowerShell note: Upar backtick `` ` `` line continuation ke liye hai. Agar masla ho to poori command ek hi line mein likh dein (har `` ` `` hata kar). CMD mein continuation ke liye `^` use hota hai.

Ek line wali (copy-paste safe) version:

```
docker run -d --name waha --restart always -p 3000:3000 -e WHATSAPP_DEFAULT_ENGINE=WEBJS -e
WAHA_API_KEY=4c4662668dff4e48997492b11d4e7754 -v waha_sessions:/app/.sessions devlikeapro/waha
```

5. (Alternative) docker-compose se chalayein

Agar aap config file mein rakhna chahte hain to ek file `docker-compose.yml` banayein:

```
services:
  waha:
    image: devlikeapro/waha
    container_name: waha
    restart: always
    ports:
      - "3000:3000"
    environment:
      - WHATSAPP_DEFAULT_ENGINE=WEBJS
      - WAHA_API_KEY=4c4662668dff4e48997492b11d4e7754
      - WAHA_PRINT_QR=false
    volumes:
      - waha_sessions:/app/.sessions

volumes:
  waha_sessions:
```

Phir us folder mein:

```
docker compose up -d
```

6. Container chal raha hai confirm karein

```
docker ps
docker logs -f waha
```

`docker ps` mein `waha` ka status `Up` hona chahiye. Logs band karne ke liye `Ctrl + C`.

7. WAHA Dashboard aur Swagger kholain

Browser mein jayein:

URL	Kya hai
`http://localhost:3000/dashboard`	Web dashboard — sessions, QR, status
`http://localhost:3000/`	Swagger API docs — testing ke liye

> Dashboard ya Swagger login API key maange to wahi key dein: `4c4662668dff4e48997492b11d4e7754`

8. "default" session banayein aur QR scan karein

EasySchool session ka naam `default` use karti hai. Do tareeqe:

Tareeqa A — Dashboard se (aasan)

1. `http://localhost:3000/dashboard` kholain
2. "Sessions" → session `default` ko **Start** karein (agar na ho to naya banayein, naam `default`)
3. QR code screen par aayega
4. Apne phone par WhatsApp → **Settings** → **Linked Devices** → **Link a Device** → QR scan karein

5. 5 Status `WORKING` ho jaye to ready hai ✓

Tareeqa B — API se (Swagger / PowerShell)

```
# 1) Session start karein
curl -X POST http://localhost:3000/api/sessions/default/start `
  -H "X-Api-Key: 4c4662668dff4e48997492b11d4e7754"

# 2) QR code haasil karein (browser mein bhi khol sakte hain)
#    http://localhost:3000/api/default/auth/qr  (X-Api-Key header ke sath)

# 3) Status check karein — "WORKING" aana chahiye
curl http://localhost:3000/api/sessions/default `
  -H "X-Api-Key: 4c4662668dff4e48997492b11d4e7754"
```

> **WEBJS pehli dafa:** Engine Chromium download/launch karti hai, is liye QR aane mein 20–60 second lag sakte hain. Sabar karein. Agar QR expire ho jaye to refresh karein.

9. EasySchool ko WAHA se connect karein

File: `School.API/appsettings.json`. Aapki app currently `Wasender` section parhti hai (historical naam) aur `Waha` section ko fallback rakhti hai. Local WAHA point karne ke liye `BaseUrl` ko localhost par set karein:

```
"Wasender": {
  "BaseUrl": "http://localhost:3000",
  "Session": "default",
  "CountryCode": "92",
  "ApiKey": "4c4662668dff4e48997492b11d4e7754"
}
```

Setting	Value	Wajah
BaseUrl	`http://localhost:3000`	Local Docker WAHA
Session	`default`	Step 8 wala session
CountryCode	`92`	Pakistan — number ko `92xxxxxxxxxx@c.us` banata hai
ApiKey	(WAHA_API_KEY jaisa)	Header `X-Api-Key` mein jata hai

> **Important:** `appsettings.json` ki `ApiKey` aur container ka `WAHA_API_KEY` **bilkul same** hone chahiyein, warna `401 Unauthorized` aayega.

10. Test message bhejein

Direct WAHA test (apna number daalein, `92` ke sath, bina `0` ke):

```
curl -X POST http://localhost:3000/api/sendText `
  -H "X-Api-Key: 4c4662668dff4e48997492b11d4e7754" `
  -H "Content-Type: application/json" `
  -d '{ "session": "default", "chatId": "923001234567@c.us", "text": "EasySchool WAHA test ✅" }'
```

Agar WhatsApp par message aa gaya to setup mukammal hai. Ab EasySchool API (`dotnet run --project School.API`) ke through SMS/WhatsApp features kaam karenge.

11. Container manage karne ke commands

Kaam	Command
Status dekhein	`docker ps`
Logs dekhein	`docker logs -f waha`
Stop karein	`docker stop waha`
Start karein	`docker start waha`
Restart	`docker restart waha`
Remove (session volume rahega)	`docker rm -f waha`
Session data delete (dobara QR)	`docker volume rm waha_sessions`

12. Troubleshooting

Masla	Hal
`401 Unauthorized`	API key mismatch — container ka `WAHA_API_KEY` aur appsettings ki `ApiKey` same karein. Header `X-Api-Key` bhejna zaroori.
QR nahi aa raha	WEBJS Chromium load ho rahi hai — 1 min wait, phir `docker logs waha` dekhein. RAM kam ho to dusri heavy apps band karein.
Session `FAILED` / `STOPPED`	Session stop karke dobara start karein. Zaroorat par `docker volume rm waha_sessions` kar ke fresh QR scan.
Message nahi ja raha	Session status `WORKING` confirm karein. Number format `92xxxxxxxxxx@c.us` ho (leading 0 nahi).
Port 3000 already in use	Dusra port: `-p 3001:3000` aur `BaseUrl` ko `http://localhost:3001` karein.
PC restart ke baad band	`--restart always` diya ho to khud start hoga. Docker Desktop "Start on login" enable rakhein.
Phone disconnect ho jata hai	Jis phone se QR scan kiya us mein internet on rahe; WhatsApp ko background mein band na karein.

Production tip: Live server par `BaseUrl` ko `https://waha-messaging.dev.crymzee.com` (ya aapka domain) rakhein aur localhost sirf development ke liye use karein.